

Aaron Broukhim

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Experience

Sep 2022 to June 2022	University of California San Diego - Graduate Researcher, PhD in CSE (Advisor: Nadir Weibel) <i>Thesis: preference-based reinforcement learning for generative speech</i> - Developed a fine-tuning method to condition a duplex model's audio output on internal uncertainty signals (semantic entropy, log-probability, hidden-state probes) without added inference latency. - Ran a between-subjects cross-modality preference study showing that text and audio judgments of identical semantic content diverge; modeled rater bias with linear mixed-effects models and validated LLM-as-judge for the audio preferences. - Collected a large-scale LLM-output preference dataset ($n \approx 3,000$; 18K+ pairwise judgments via Prolific/Qualtrics) for a study on content-moderation preferences across political affiliation. - Built a Mixture-of-Experts model with weight averaging, exploring compute/performance trade-offs.
June 2026 to Sept 2026	Adobe - Machine Learning Engineering Intern - Shipped an end-to-end query autocomplete system into a production LLM-agent platform (fuzzy-trie retrieval, freq$\times$cosine ranking, fine-tuned T5-220M fallback); cut latency $\sim 10\times$ (906ms $\rightarrow$ 93ms) and aligned generation to accept/dismiss signals via offline KTO. - Trained linear probes on frozen T5 activations for accuracy, toxicity, PII, and out-of-scope gating (AUROC 0.88–0.97), enabling real-time filtering without a second model.
Jan 2022 to Aug 2022	Virufy - Associate Audio Machine Learning Engineer Fine-tuned audio classifiers via transfer learning (TensorFlow, SK-Learn) for cough-based detection; ran feature/architecture search on SageMaker and peer-reviewed ML research.
July 2021 to Aug 2022	University of California San Diego - Research Assistant Built logistic-regression and word-embedding models for hate-speech detection on Twitter; scraped data with Selenium; shipped a React/Node/Flask + AWS/SQL tool for non-experts to audit faulty ML systems.

Education

June 2027	University of California San Diego - Ph.D. in Computer Science & Engineering
June 2021	University of California San Diego - B.S. in Cognitive Science: Machine Learning

Selected Publications ([Google Scholar](#))

Accepted or Published	<u>Broukhim</u> et al., Same Words, Different Judgments: How Preferences Vary Across Modalities, NeurIPS 2026 <u>Kaufman</u> , <u>Broukhim</u> et al., WARNING This Contains Misinformation: The Effect of Cognitive Factors, Beliefs, and Personality on Misinformation Warning Tag Attitudes, CSCW 2025 <u>Kaufman</u> , <u>Broukhim</u> et al., What Did My Car Say? Impact of Autonomous Vehicle Explanation Errors and Driving Context On Comfort, Reliance, Satisfaction, and Driving Confidence, CHI 2025
In Prep. or Under Review	<u>Broukhim</u> et al., Mental Health Red-Teaming of LLMs for Vulnerable Low Resource Patients, <i>in preparation</i> <u>Broukhim</u> et al., Aligning Perceived Confidence to Generated Speech, <i>in preparation</i> <u>Broukhim</u> et al., Synthetic Social Emotion Dysregulation Evaluation Dataset and Models, <i>in preparation</i> <u>Broukhim</u> et al., Preference-Based Learning in Audio Applications: A Systematic Analysis, <i>under review</i> , <i>ACM Computing Surveys</i>

Skills

Tech Skills	Languages: Python, SQL, Java, C++, C, JavaScript, HTML/CSS ML/Frameworks: PyTorch, TensorFlow, Keras, SK-Learn, HuggingFace, NumPy, Pandas Infra/Tools: AWS, SageMaker, Git, Firebase, MySQL, React/Node/Flask, Selenium, Kubernetes
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